

ELTE IK - Computer Science BSc

Final Exam Topics

2.1 Differential equations

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1 The initial value problem

Differential equation

$0 < n \in \mathbb{N}$, $I \subset \mathbb{R}$ open interval,
 $\Omega := I_1 \times \dots \times I_n \subset \mathbb{R}^n$, where $I_1, \dots, I_n \subset \mathbb{R}$ open interval
 $f : I \times \Omega \rightarrow \mathbb{R}^n$, $f \in C$

Determine the function $\varphi \in I \rightarrow \Omega$ such that:

- D_φ open interval
- $\varphi \in D$
- $\varphi'(x) = f(x, \varphi(x)) \quad (x \in D_\varphi)$

We call this task a differential equation.

Initial value problem

If, in addition to the above, the following are also given: $\tau \in I$, and $\xi \in \Omega$
And the function φ additionally satisfies:

- $\tau \in D_\varphi$ and $\varphi(\tau) = \xi$

Then we call it an initial value problem (Cauchy problem).

2 Linear and higher-order linear differential equations

2.1 Linear differential equations

Definition

A linear differential equation is a differential equation for which:

$n = 1$, $I, I_1 \subset \mathbb{R}$ open intervals, $f : I \times I_1 \rightarrow \mathbb{R}$, where

$g, h : I \rightarrow \mathbb{R}$, $g, h \in C$, $I_1 := \mathbb{R}$ and

$f(x, y) := g(x) \cdot y + h(x) \quad (x \in I, y \in I_1 = \mathbb{R})$

$\Rightarrow \varphi'(x) = f(x, \varphi(x)) = g(x) \cdot \varphi(x) + h(x) \quad (x \in D_\varphi)$

Homogeneity

The linear differential equation is homogeneous if $h \equiv 0$ (otherwise inhomogeneous)

Initial value problem

- Every initial value problem relating to a linear differential equation is solvable and for $\forall \varphi, \psi$ solutions: $\varphi(t) = \psi(t) \quad (t \in D_\varphi \cap D_\psi)$

- Every solution of a homogeneous linear differential equation ($\varphi : I \rightarrow \mathbb{R}$) has the following form:
 $c\varphi_0$, where
 $c \in \mathbb{R}$ and $\varphi_0(t) = e^{G(t)}$ ($G : I \rightarrow \mathbb{R}$, $G \in D$, and $G' = g$)
- Method of variation of constants:
 $\exists m : I \rightarrow \mathbb{R}$, $m \in D : m \cdot \varphi_0$ is a solution of the (inhomogeneous) linear differential equation
- Particular solution:
 $M := \{\varphi : I \rightarrow \mathbb{R} : \varphi'(t) = g(t) \cdot \varphi(t) + h(t) \ (t \in I)\}$
 $M_h := \{\varphi : I \rightarrow \mathbb{R} : \varphi'(t) = g(t) \cdot \varphi(t) \ (t \in I)\}$
 $\Rightarrow \forall \psi \in M : M = \psi + M_h = \{\varphi + \psi : \varphi \in M_h\}$
 (And here ψ , based on the above, can be written in the form $m \cdot \varphi_0$)
- Example: Radioactive decay:
 $m_0 > 0$ - initial amount of material
 $m \in \mathbb{R} \rightarrow \mathbb{R}$ - mass-time function, where
 $m(t)$ - the amount of material present
 $m \in D \Rightarrow \frac{m(t) - m(t + \Delta t)}{\Delta t}$ ($\Delta t \neq 0$) - average decay rate
 $\frac{m(t) - m(t + \Delta t)}{\Delta t} \xrightarrow{\Delta t \rightarrow 0} -m'(t)$, which based on observation is $\approx m(t)$
 that is:
 $m'(t) = -\alpha \cdot m(t) \ (t \in \mathbb{R}, 0 < \alpha \in \mathbb{R})$
 $m(0) = m_0$

Homogeneous linear differential equation (initial value problem):

$$\begin{aligned}
 g &\equiv -\alpha, \ \tau := 0, \ \xi := m_0 \\
 \Rightarrow G(t) &= -\alpha t \ (t \in \mathbb{R}) \Rightarrow \varphi_0(t) = e^{-\alpha t} \ (t \in \mathbb{R}) \\
 \Rightarrow \exists c \in \mathbb{R} : m(t) &= c \cdot e^{-\alpha t} \ (t \in \mathbb{R}), \text{ where} \\
 m(0) = c = m_0 &\Rightarrow m(t) = m_0 e^{-\alpha t} \ (t \in \mathbb{R}) \\
 \text{If } T \in \mathbb{R} : m(T) &= \frac{m_0}{2} \text{ (half-life)} \\
 \Rightarrow \frac{m_0}{2} &= m_0 e^{-\alpha T} \Rightarrow \frac{1}{2} = e^{-\alpha T} \Rightarrow e^{\alpha T} = 2 \\
 \Rightarrow T &= \frac{\ln(2)}{\alpha}
 \end{aligned}$$

2.2 Higher-order linear differential equations

Definition

$0 < n \in \mathbb{N}$, $I \subset \mathbb{R}$ open, $a_0, \dots, a_{n-1} : I \rightarrow \mathbb{R}$ continuous and $c : I \rightarrow \mathbb{R}$ continuous.

Let us look for a function $\varphi \in I \rightarrow \mathbb{K}$ such that:

- $\varphi \in D^n$
- D_φ open interval
- $\varphi^{(n)}(x) + \sum_{k=0}^{n-1} a_k(x) \cdot \varphi^{(k)}(x) = c(x) \ (x \in D_\varphi)$

We call this an n -th order linear differential equation. (For $n = 1$, a linear differential equation.)

If additionally:

$\tau \in I$, $\xi_0, \dots, \xi_{n-1} \in \mathbb{K}$ and

- $\tau \in D_\varphi$ and $\varphi^{(k)}(\tau) = \xi_k \ (k = 0 \dots n-1)$

Then we speak of an initial value problem.

Homogeneity

If $c(x) = 0$ we speak of a homogeneous n -th order linear differential equation. So the solution sets of the homogeneous and inhomogeneous equations:

$$M_h := \{\varphi : I \rightarrow \mathbb{K} : \varphi \in D^n, \ \varphi^{(n)} + \sum_{k=0}^{n-1} a_k \cdot \varphi^{(k)} = 0\}$$

$$M := \{\varphi : I \rightarrow \mathbb{K} : \varphi \in D^n, \varphi^{(n)} + \sum_{k=0}^{n-1} a_k \cdot \varphi^{(k)} = c\}$$

(Here M_h is an n -dimensional linear space, so some $\varphi_1, \dots, \varphi_n \in M_h$ form a basis, also called a fundamental system.)

Constant-coefficient case

In this case $a_0, \dots, a_{n-1} \in \mathbb{R}$

- Role of the characteristic polynomial

Let $P(t) := t^n + \sum_{k=0}^{n-1} a_k t^k$ ($t \in \mathbb{K}$) be the characteristic polynomial and

$$\varphi_\lambda(x) := e^{\lambda x} \quad (x \in \mathbb{R}, \lambda \in \mathbb{K})$$

Then: $\varphi_\lambda \in M_h \iff P(\lambda) = 0$

Moreover, if λ is an r -fold root of P , and

$$\varphi_{\lambda,j}(x) := x^j e^{\lambda x} \quad (j = 0..r-1, x \in \mathbb{R}), \text{ then: } \varphi_{\lambda,j} \in M_h \iff \varphi_{\lambda,j}^{(n)} + \sum_{k=0}^{n-1} a_k \varphi_{\lambda,j}^{(k)} = 0$$

that is $P(\lambda)^{(j)} = 0$ ($j = 0..r-1$)

- Real solutions

Let $\lambda = u + iv$ ($u, v \in \mathbb{R}, v \neq 0, i^2 = -1$)

$\Rightarrow$ the functions $x \mapsto x^j e^{ux} \cos(vx)$, and $x \mapsto x^j e^{ux} \sin(vx)$ form a real fundamental system (basis) (in M_h)

Example: Oscillations

Describe the motion of a point mass of mass m performing oscillatory motion along a straight line around a fixed point, if we know the position it occupied at the start of the observation and its velocity at that time!

$\varphi \in \mathbb{R} \rightarrow \mathbb{R}, \varphi \in D^2$: displacement-time function

$m > 0$: mass

$F \in \mathbb{R} \rightarrow \mathbb{R}$: displacing force

$\alpha > 0$: restoring force, proportional to φ

$\beta \geq 0$: damping force, proportional to the velocity.

$\implies$ (based on Newton's law of motion):

$$m \cdot \varphi'' = F - \alpha \varphi - \beta \varphi'$$

$$\varphi(0) = s_0, \varphi'(0) = s'_0$$

Second-order linear differential equation (initial value problem)

$$\text{Written in standard form: } \varphi'' + \frac{\beta}{m} \varphi' + \frac{\alpha}{m} \varphi = \frac{F}{m}$$

Consider the periodic external constraint as a forced oscillation, when:

$$\frac{F(x)}{m} = A \sin(\omega x) \quad [A > 0 \text{ (amplitude)}, \omega > 0 \text{ (forcing frequency)}]$$

Then $\omega_0 := \sqrt{\frac{\beta}{m}}$ - natural frequency

$$\text{and } \varphi''(x) + \omega_0^2 \varphi(x) = A \sin(\omega x)$$

whose characteristic polynomial is: $P(t) = t^2 + \omega_0^2$ ($t \in \mathbb{R}$)

Its solutions: $\lambda = \pm \omega_0 i$

We saw earlier that if $\lambda = u + iv$ then the functions $x \mapsto x^j e^{ux} \cos(vx)$, and $x \mapsto x^j e^{ux} \sin(vx)$ form a real fundamental system (basis) (in M_h). So $\varphi(x) = c_1 \cos(\omega_0 x) + c_2 \sin(\omega_0 x)$ can be written in a form which, using a phase angle, can be rewritten as: $d \cdot \sin(\omega_0 x + \delta)$ ($d = \sqrt{c_1^2 + c_2^2}, \delta \in \mathbb{R}$). So: $M_h = \{d \cdot \sin(\omega_0 x + \delta)\}$

Then we can already easily give a particular solution:

- for $\omega \neq \omega_0$ the particular solution:

$$x \rightarrow q \cdot \sin(\omega x)$$

And $q = \frac{A}{\omega_0^2 - \omega^2}$ satisfies the equation $-q\omega^2 \sin(\omega x) + \omega_0^2 q \cdot \sin(\omega x) = A \sin(\omega x)$. So:

$\varphi(x) = d \cdot \sin(\omega_0 x + \delta) + \frac{A}{\omega_0^2 - \omega^2} \sin(\omega x)$ the solution is the sum of two harmonic oscillations.

- for $\omega = \omega_0$ (resonance) the particular solution:

$$x \rightarrow qx \cdot \cos(\omega x)$$

And $q = \frac{-A}{2\omega}$ satisfies the equation $-2q\omega \cdot \sin(\omega x) - q\omega^2 x \cdot \cos(\omega x) + \omega^2 qx \cdot \cos(\omega x) = A \sin(\omega x)$.

So:

$\varphi(x) = d \cdot \sin(\omega x + \delta) - \frac{A}{2\omega} x \cdot \cos(\omega x)$ the solution is the sum of a harmonic and an aperiodic oscillation.

(In this case, as time (x) passes, the value of φ grows. In certain models this causes the “disintegration of the system”.)